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Undergraduate Thesis (Design)

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Content Checklist

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Abstract

Clinical decision-making is inherently longitudinal — clinicians must synthesize evidence across repeated hospitalizations, diagnostic tests, and evolving treatment responses spanning months or years. However, current benchmarks for medical large language model (LLM) predominantly evaluate single-encounter behavior, focusing on model knowledge, tool-use accuracy and fact retrieval rather than the cross-visit temporal integration that real clinical practice demands. To address this gap, this thesis introduces the decision-making component of LongMedBench, a benchmark constructed from real-world Electronic Health Record (EHR) data (MIMIC-IV) that evaluates LLM agents on long-horizon clinical decision-making. We design a three-tier memory architecture operating at three granularities — Note Memory (visit-level summaries), Event Memory (fine-grained structured event sequences), and Contextual Memory (immediate visit context) — and systematically compare memory augmentation strategies including naive long-context, retrieval-augmented generation (RAG), and agent memory systems (Mem0). Through experiments on GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo, we find that decision-making accuracy across all models remains modest, and critically, that memory augmentation architectures provide no significant improvement over the naive long-context baseline. Decision accuracy depends primarily on the immediate clinical context, with historical information contributing negligible marginal value regardless of memory granularity or retrieval strategy. This central finding reveals a retrieval-reasoning gap: current architectures can locate relevant historical facts but cannot effectively integrate them into proactive clinical planning. We discuss the implications of this gap for medical AI agent design and propose directions for memory architectures that go beyond retrieval to support structured longitudinal reasoning.

Keywords: Medical AI Agents, Clinical Decision-Making, Electronic Health Records, Long-Horizon Reasoning, Retrieval-Augmented Generation, Agent Memory Systems

Content

Chapter 1. Introduction

1.1 The Long-Horizon Challenge in Clinical Decision-Making

Clinical practice is fundamentally longitudinal. Patients with chronic conditions accumulate medical histories spanning years and dozens of hospitalizations, and at each encounter clinicians must synthesize evidence across prior diagnostic tests, imaging studies, medication adjustments, and evolving treatment responses [1]. This capacity for cross-visit temporal integration is central to clinical reasoning, and it represents a capability that medical AI must acquire to be clinically useful.

As large language models (LLMs) transition from static question-answering to autonomous medical agents, their ability to navigate longitudinal clinical trajectories has become a critical benchmark for clinical readiness. Yet existing evaluation benchmarks assess single-encounter behavior: tool-use accuracy in a simulated visit, diagnosis from a single symptom set, or fact retrieval from a single discharge summary. These evaluations, while valuable, do not measure the cross-visit temporal integration that real clinical practice demands.

This thesis presents the decision-making component of **LongMedBench**, a benchmark constructed from MIMIC-IV [2] EHR data that evaluates LLM agents on long-horizon clinical decision-making. Our work bridges two previously disconnected lines of research: the design of long-horizon interaction benchmarks from the general AI domain, and the evaluation of clinical reasoning from real-world EHR data. Through a three-tier memory architecture and systematic ablation of memory strategies, we investigate a central question: given long-context windows, RAG, and agent memory systems, how well can contemporary LLMs integrate longitudinal clinical history to inform next-step decisions?

1.2 Medical Benchmarks for Clinical AI

Medical AI evaluation has evolved from static QA to interactive agent settings, as shown in Table 1.1. Early benchmarks including MedBench [3] and PromptCBLUE [4]

Table 1.1 Comparison of medical benchmarks for clinical AI. *Legend:* ✓ = capability present. EHR = real electronic health record data. Agent = interactive agent-based evaluation. Memory Study = systematic memory architecture ablation.

Benchmark	EHR	Multi-Visit	Agent	Decision-Making	Temporal	Memory Study	Year	Data Source
MedBench [3]							2023	Exams
PromptCBLUE [4]							2023	Exams
EHRSQL [5]	✓						2022	MIMIC-IV
MedAlign [6]	✓			✓			2024	STARR
MedAgentBench [7]	✓		✓	✓			2025	STARR
AgentClinic [8]	✓		✓	✓			2025	Simulated
ER-REASON [9]	✓		✓	✓			2025	MIMIC-IV
FHIR-AgentBench [10]	✓		✓		✓		2025	MIMIC-IV
MedAgentGym [11]	✓		✓	✓			2025	Multiple
TIMER [12]	✓	✓	✓	✓	✓		2025	STARR
EHR-Bench [13]	✓	✓	✓	✓			2025	MIMIC-IV
LongMedBench	✓	✓	✓	✓	✓	✓	2026	MIMIC-IV

etc., tested LLm's knowledge recall from medical licensing exams. Second-generation work introduced real clinical data: EHRSQL [5] pioneered text-to-SQL on structured EHR; FHIR-AgentBench [10] extended this to interoperable FHIR standards; MedAlign [6] provided clinician-generated instruction-following tasks grounded in EHR data.

The above two categories assess the static capabilities of the model; however, real-world medical tasks are complicated. A series of agent-based evaluation frameworks have been designed, wherein researchers construct simulated environments that enable LLMs to interactively complete complex tasks. MedAgentBench [7] introduced a virtual EHR environment with 300 clinician-written tasks. AgentClinic [8] added multimodal clinical encounters. ER-REASON [9] focused on emergency room reasoning with structured subtasks. MedAgentGym [11] provided a scalable training environment for biomedical reasoning.

Despite this progress, a critical gap persists: all current medical benchmarks evaluate single-encounter behavior. Real clinical care spans dozens of visits over years, and a patient's current presentation cannot be understood in isolation from their clinical trajectory. Meanwhile, no existing medical benchmark systematically compares memory architectures of the AI agent for clinical decision-making.

1.3 Long-Horizon Benchmarks in the General Domain

In parallel with medical AI, a rich set of long-horizon benchmarks was presented in the general community of AI agents, as shown in Table 1.2. Their task design principles inspire the construction of longitudinal clinical benchmarks, which includes multi-turn interaction, cross-session memory, progressive information revelation.

Long-Context Retrieval. RULER [14] systematically evaluates the effective context window of LLMs through synthetic tasks like needle-in-a-haystack, variable tracking, etc., at controlled lengths up to 128K tokens. NarrativeQA [15] tests reading comprehension over full-length books and movie scripts, requiring models to answer questions that depend on information distributed across long documents. HotpotQA [16] requires reasoning across multiple documents to answer questions.

Multi-Hop and Multi-Turn Benchmarks. Beyond basic retrieval challenges, to test whether models can reason across temporally structured information, LoCoMo [17] and LongMemEval [18] uses simulated multi-session dialogues to evaluate whether models can recall and reason about information shared in earlier sessions. Both introduce the concept of multi-hop reasoning, which parallels the clinical need to connect evidence across visits.

Interactive Agent Benchmarks. TRIP-Bench [19] evaluates agents on executing long-horizon interactive tasks in real-world scenarios, where agents must plan, execute, and adapt across extended interactions. AMA-Bench [20] benchmarks long-horizon memory for agentic applications, measuring how well agents retain and leverage information across prolonged task sequences. In both cases, the agent navigates a sequence of steps informed by all prior history, and evaluation centers on whether historical information is effectively integrated into next-step decisions. This paradigm inspires LongMedBench's clinical decision-making setting.

Table 1.2 Long-horizon and long-context benchmarks in the general domain. *Legend:* ✓= capability present. Long Context = requires processing contexts exceeding typical window sizes. Interactive = involves multi-step agent-environment interaction.

Benchmark	Multi-Turn	Long Context	Interactive	Reasoning	Key Design	Year
NarrativeQA [15]		✓			Full-book comprehension	2017
HotpotQA [16]				✓	Multi-doc reasoning	2018
RULER [14]		✓			Synthetic context stress test	2024
LoCoMo [17]	✓	✓	✓	✓	Multi-session dialogues	2024
LongMemEval [18]	✓	✓	✓	✓	Long-term chat memory	2024
TRIP-Bench [19]	✓	✓	✓	✓	Real-world interactive planning	2026
AMA-Bench [20]	✓	✓	✓	✓	Memory for agentic applications	2026

1.4 Memory-Augmented Architectures for Long-Horizon Interaction

To address the challenges exposed by long-horizon benchmarks, a parallel line of research has developed memory-augmented architectures that extend LLM capabilities beyond their native context windows.

Retrieval-Augmented Generation (RAG) The foundational RAG paradigm [21] augments LLM inference by retrieving relevant documents from an external knowledge base and prepending them to the context window, enabling scale beyond the context window at the cost of retrieval quality.

Agent Memory Systems Among agent-native memory frameworks, we adopt Mem0 [22] as the representative agent memory system in our experiments. Mem0 provides a persistent, evolvable memory layer with add, search, and update operations, occupying a practical middle ground: it is more persistent and agent-driven than retrieval-only RAG, yet simpler and more mature than research-stage frameworks, with a clean API well-suited for systematic ablation across memory granularities. MemGPT [23] introduced an OS-inspired memory hierarchy with working and archival storage. More recent frameworks have pushed further: A-MEM [24] explicitly models agentic memory decisions, where the agent autonomously determines what to retain; EverMemOS [25] proposes a self-organizing memory operating system for structured long-horizon reasoning; and Memory as Action [26] reframes memory management as an action the agent takes, integrating it into the decision loop rather than treating it as preprocessing.

Table 1.3 summarizes these frameworks chronologically. Despite their architectural diversity, a common assumption underlies them: that better memory retrieval leads to better task performance. Whether this assumption holds for proactive clinical decision-making is a key question that LongMedBench is designed to answer.

Table 1.3 Memory-augmented architectures for LLM agents. *Legend:* ✓= yes. Persistent = memory state survives across sessions/interactions.

Framework	Strategy	Key Mechanism	Persist.	Year	Innovation
RAG [21]	Retrieve-then-generate	Embed + Retrieve + Augment	Ext.	2020	Separates storage & inference
MemGPT [23]	Hierarchical memory	Core + Archival paging	✓	2023	OS-inspired management
Mem0 [22]	Agent memory layer	Add + Search + Update	✓	2024	Multi-session persistence
A-MEM [24]	Agentic memory	Autonomous retention	✓	2025	Agent-driven memory
Memory as Action [26]	Memory-as-decision	Context curation loop	✓	2025	Memory in decision loop
EverMemOS [25]	Self-organizing OS	Structured long-horizon	✓	2026	Self-organizing memory

1.5 LongMedBench

LongMedBench is positioned at the intersection of these two bodies of work. On one hand, it inherits the domain grounding of medical benchmarks. On the other hand, it adopts the task design methodology of general long-horizon benchmarks.

1. **Innovation in medical AI:** LongMedBench is the first benchmark to combine real EHR data with interactive agent evaluation and systematic memory architecture ablation. No prior medical benchmark (Table 1.1) has systematically compared how different memory strategies—naive long-context, retrieval-augmented, and agent-managed memory—affect clinical decision-making.

2. **Innovation in long-horizon AI:** LongMedBench is the first benchmark to apply long-horizon interaction task design (as developed in NarrativeQA, RULER, LoCoMo, TRIP-Bench) to real-world clinical data. Unlike general-domain benchmarks that use synthetic or curated text, LongMedBench tests long-horizon reasoning on noisy, real-world clinical trajectories with temporal structure and clinical consequence.

1.6 Research Questions

Specifically, this thesis investigates three questions:

1. **Memory granularity:** Does the format of historical information—structured event-level data vs. narrative note-level summaries—affect clinical decision quality?

2. **Memory architecture:** Do RAG and agent memory systems (Mem0) improve longitudinal decision-making relative to providing the full history in context?

3. **Historical depth:** Does decision performance plateau, improve, or degrade as more historical visits are included?

1.7 Contributions

Within the broader LongMedBench project (whose factual QA and temporal reasoning tasks are undertaken by collaborators), this thesis makes four contributions:

1. **EHR-to-Agent Data Pipeline.** A reproducible pipeline transforming raw MIMIC-IV records into longitudinal event streams for interactive agent evaluation, yielding 355 patients and 6,999 hospitalization visits (mean 19.72 visits/patient, 44.91 medical events/visit).

2. **Three-Tier Memory Architecture.** A memory framework operating at three granularities—Note Memory (coarse-grained visit summaries), Event Memory (fine-grained structured event sequences), and Contextual Memory (immediate visit context)—enabling systematic ablation of memory granularity effects.

3. **Long-Horizon Decision-Making Task.** An interactive evaluation protocol where the agent receives a partial patient trajectory through controlled memory and must predict the next clinical action from a predefined action space, with ground truth extracted from actual clinical records.

4. **Empirical Finding.** Through experiments across GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo comparing Naive Long-Context, RAG, and Mem0 strategies, we find that memory augmentation provides no significant improvement for longitudinal clinical decision-making. Decision accuracy depends primarily on immediate context, revealing a retrieval-reasoning gap for longitudinal clinical decision-making.

1.8 Scope and Structure

This thesis focuses exclusively on Long-Horizon Decision-Making Tasks. The Fact-based QA and Temporal Reasoning tasks are covered in collaborators' separate theses.

Chapter 2 presents the data processing pipeline, three-tier memory architecture, decision-making task design, experimental setup, and results. Chapter 3 discusses the implications of our findings, acknowledges limitations, and outlines future research directions.

Chapter 2. Methodology and Experiments

This chapter presents the LongMedBench decision-making pipeline: from raw EHR data to interactive agent evaluation. Section 2.1 describes the data processing pipeline. Section 2.2 presents the three-tier memory architecture. Section 2.3 formulates the T3-D decision-making task. Section 2.4 details the experimental design. Section 2.5 presents results and analysis.

2.1 EHR Data Processing Pipeline

LongMedBench is constructed from MIMIC-IV [2], a publicly available de-identified database containing comprehensive clinical data for over 100,000 patients at Beth Israel Deaconess Medical Center (2008–2019). The database includes structured tables (laboratory measurements, medication administrations, procedure records) and unstructured clinical notes (radiology reports, discharge summaries). To convert this static relational database into an interactive agent-compatible environment, we design a three-stage pipeline.

2.1.1 Patient and Event Filtering

Event-Level Filtering. We retain only laboratory results flagged as abnormal. Normal lab values carry limited clinical signal for decision-making and inflate context without benefit.

Patient-Level Filtering. We require complete admission and discharge records for all hospitalizations, and select patients with at least 15 recorded hospitalizations. This threshold yields trajectories with sufficient density for longitudinal reasoning evaluation.

Resulting Cohort. The filtered dataset comprises **355 patients** with **6,999 inpatient visits**. Mean visits per patient: 19.72 (median = 18.00, SD = 5.72). Mean medical events per visit: 44.91 (median = 25.00, SD = 70.32). This dense multi-session structure challenges models on cross-visit memory retention and temporal reasoning.

The ≥ 15 hospitalization filter creates a high-acuity, chronic-disease subpopulation—patients with the most complex conditions requiring frequent care. This is precisely the population for whom longitudinal reasoning matters most, but findings may not generalize to lower-acuity settings (discussed in Chapter 3).

2.1.2 Event Stream Construction

We transform filtered records into a unified, time-ordered event stream for each patient. A medical event is any discrete clinical occurrence, formalized as $E_i^k = \{a_i^k, t_i^k, p_i^k, o_i^k\}$, where a_i^k is the action type (imaging, lab test, medication, procedure, etc.), t_i^k is the timestamp, p_i^k is the action parameter (e.g., CT vs. X-ray for imaging), and o_i^k is the observation result (radiology report, lab value, administration confirmation).

Each visit $\mathcal{V}_i$ is bounded by an admission event E_i^{adm} and a discharge event E_i^{dis}: $\mathcal{V}_i = \{E_i^{adm}, E_i^1, E_i^2, \ldots, E_i^m, E_i^{dis}\}$. The complete patient stream is $\mathcal{S}_\mathcal{P} = \{\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_n\}$, where n is the number of visits.

Note Processing. Radiology reports are converted to structured imaging events via regular expression parsing. Discharge notes N_i are decomposed into admission information N_i^{adm} and discharge information N_i^{dis}, preserved in narrative format for use in Note Memory (Section 2.2.1) and ground-truth extraction.

2.1.3 Reproducibility

The pipeline is implemented as standardized, modular, version-controlled scripts. Each stage (extraction, event construction, task generation) is independently executable, enabling end-to-end reproducibility and adaptability to other EHR databases.

2.2 Three-Tier Memory Architecture

A central design principle is the controlled provision of historical clinical information. Rather than exposing the agent to the full patient history without control over format or retrieval, we define a three-tier memory architecture that systematically varies both the granularity and the provision strategy of historical information.

2.2.1 Note Memory (Coarse-Grained)

Note Memory consists of admission and discharge summaries (N_i^{adm}, N_i^{dis}) for each historical visit, presented as structured natural language text (200–500 words per visit) concatenated in chronological order. It is compact and preserves clinical narrative structure, but is inherently lossy: summarization discards fine-grained temporal and event-level details.

2.2.2 Event Memory (Fine-Grained)

Event Memory contains the complete structured event sequence for each historical visit—every lab test, medication, imaging study, and procedure with full action type, timestamp, parameters, and results. Events are presented as time-ordered JSON objects grouped by visit. Event Memory is information-complete, but its volume (mean 900 events across 20 visits) approaches or exceeds typical LLM context windows, creating a natural stress test for both long-context capabilities and retrieval strategies.

2.2.3 Contextual Memory (Immediate)

Contextual Memory is limited to the current visit's partial event history (all events from admission up to the decision point). It represents the baseline information available at any decision point and is always provided to the agent. Historical memory (Note or Event) is the experimental variable, enabling isolation of the marginal contribution of history beyond the immediate clinical context.

2.2.4 Memory Provision Strategies

Three strategies for providing historical memory are compared:

Naive Long-Context (Baseline). All historical memory is provided in full within the model's context window, testing native long-context attention without external retrieval assistance.

RAG. Historical memory is externally indexed. At decision time, the current clinical context serves as a retrieval query; top-k relevant chunks are retrieved and concatenated with the current context. This tests whether selective retrieval overcomes lost-in-the-middle effects.

Mem0 (Agent Memory). We integrate the Mem0 framework [22], providing a persistent, evolvable memory store. The agent can add, search, and update memory across a patient's trajectory, actively managing what is stored and retrieved. This tests whether agentic memory management outperforms passive retrieval.

2.3 Long-Horizon Decision-Making Task (T3-D)

The T3-D task evaluates an LLM agent's ability to make proactive clinical decisions based on partial patient trajectories. Unlike static benchmarks that present complete histories for classification, T3-D simulates interactive decision-making: the agent receives information incrementally and must determine the next action at each decision point.

2.3.1 Task Formulation

At each decision point, the agent receives: (1) a system prompt establishing clinical role and task instructions; (2) Contextual Memory (current visit's partial history); (3) Historical Memory (prior visits, at controlled granularity and provision strategy); and (4) a decision query directing the agent to predict the next clinical action. The agent outputs a predicted action $\hat{a}$ from the clinical action space $\mathcal{A}$, which is compared against ground-truth action a^* extracted from actual records.

2.3.2 Action Space and Ground Truth

The action space $\mathcal{A}$ includes: Imaging (CT, X-ray, MRI, Ultrasound), Laboratory Tests (CBC, metabolic panels, cultures), Medications (drug prescriptions with name/dose/route), Procedures (intubation, central lines), Consultations (specialty referrals), Admission/Discharge actions, and Transfers (between care units).

Ground-truth actions are extracted through structured field extraction from MIMIC-IV tables and regular expression parsing of discharge notes. Extracted actions are validated for temporal coherence (e.g., medication order before administration, discharge terminating the visit).

Decision points are sampled to cover diverse clinical scenarios: early in the visit (post-admission), mid-visit (during active treatment), and late in the visit (near discharge). Each patient contributes multiple decision points across their visits, yielding approximately 9,600 decision instances.

2.3.3 Evaluation Metrics

Exact Match (EM): Proportion of instances where predicted action matches ground truth exactly (action type + parameter). **Hit-Any (HA):** Proportion where predicted action type matches ground truth type, regardless of parameter (e.g., "CT Chest" vs. "X-ray Chest" both count as imaging). Per-category metrics are reported disaggregated by action type.

These metrics capture complementary aspects: EM is strict, penalizing clinically reasonable alternatives; HA relaxes parameter specificity, acknowledging that action category may be more informative.

2.4 Experimental Setup

2.4.1 Models

We evaluate three state-of-the-art LLMs: GPT-5-mini (400K context window), DeepSeek-v3.2 (128K context, with and without chain-of-thought reasoning mode), and Qwen-Turbo (128K context, baseline for ablation studies). All are evaluated through their respective APIs with standardized prompts.

2.4.2 Experimental Conditions

We design a factorial ablation varying four factors:

1. **Memory Granularity:** Note Memory vs. Event Memory.
2. **Provision Strategy:** Naive Long-Context vs. RAG vs. Mem0.
3. **Historical Depth (n):** Number of injected historical visits: $n \in \{0, 1, 2, 3, \infty\}$. $n = 0$ provides Contextual Memory only (no history) as baseline.
4. **Reasoning Mode:** Standard vs. chain-of-thought (DeepSeek-v3.2 only).

Qwen-Turbo serves as the primary ablation vehicle; key conditions are replicated on GPT-5-mini and DeepSeek-v3.2 for cross-model validation.

2.5 Results and Analysis

2.5.1 Overall Decision-Making Performance

Table 2.1 summarizes T3-D performance across models. Decision-making accuracy falls within a narrow band of 0.55–0.62 (Exact Match), confirming that T3-D presents a genuine challenge that current LLMs have not mastered. The modest ceiling validates the benchmark's difficulty and its capacity to differentiate future improvements.

Table 2.1 Overall T3-D performance (Exact Match).

Model	Note Mem.	Event Mem.	Context
GPT-5-mini	[TODO]	0.607	400K
DeepSeek-v3.2	[TODO]	0.607	128K
DeepSeek-v3.2-thinking	[TODO]	0.593	128K
Qwen-Turbo	[TODO]	0.604	128K

2.5.2 Memory Granularity: Note vs. Event

Table 2.2 compares performance across memory granularities. Structured Event Memory offers slight advantages over Note Memory for most models, but the difference is modest—consistent with the interpretation that for proactive clinical planning, immediate context outweighs historical granularity.

Table 2.2 Effect of memory granularity on T3-D (Naive Long-Context).

Model	Note	Event	Δ
GPT-5-mini	[TODO]	0.607	[TODO]
DeepSeek-v3.2	[TODO]	0.607	[TODO]
Qwen-Turbo	[TODO]	0.604	[TODO]

2.5.3 Memory Architecture: Naive vs. RAG vs. Mem0

This is the central experimental question. Table 2.3 presents the memory architecture ablation using Qwen-Turbo.

Finding. Neither RAG retrieval nor agent memory management (Mem0) provides significant improvement over the naive long-context baseline. In some conditions, augmentation slightly degrades performance. This negative result is the central empirical

Table 2.3 Effect of memory architecture on T3-D (Qwen-Turbo, Event Memory).

Architecture	T3-D Acc.	Δ vs. Naive	Notes
Naive (Full)	[TODO]	—	Baseline
RAG	[TODO]	[TODO]	Top-k retrieval
Mem0	[TODO]	[TODO]	Agent memory

contribution: it reveals that the bottleneck in longitudinal clinical decision-making is not retrieval (locating relevant history) but integration (understanding how historical information bears on the current decision). Current architectures optimize for retrieval quality, but our results indicate that for clinical planning, closing the gap between retrieval and reasoning is the critical challenge.

2.5.4 Historical Depth: Visit Injection Ablation

Table 2.4 shows performance as a function of injected historical visits (n). The $n = 0$ baseline (Contextual Memory only, no history) is particularly informative.

Table 2.4 Effect of historical depth on T3-D (Qwen-Turbo, Event Memory).

n (Visits)	T3-D Acc.	Interpretation
0 (Context only)	[TODO]	Baseline: no history
1	[TODO]	Single prior visit
2	[TODO]	Two prior visits
3	[TODO]	Three prior visits
∞ (All)	[TODO]	Complete history

Finding. Performance does not monotonically improve with historical depth. The $n = 0$ baseline achieves accuracy comparable to full-history conditions, indicating that decisions are driven primarily by the immediate clinical presentation. Adding history introduces noise that can offset the value of relevant historical information—consistent with the lost-in-the-middle effect [27].

2.5.5 Chain-of-Thought Reasoning

For DeepSeek-v3.2, explicit chain-of-thought reasoning reduces T3-D accuracy from 0.607 to 0.593. Qualitative analysis of reasoning traces suggests that thinking mode makes the model more conservative—it identifies information gaps and hesitates rather than committing to a decision. This highlights a tension between clinical caution and benchmark performance.

2.5.6 Cross-Model Comparison

[TODO: Brief comparison of performance across GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo. Note whether larger context windows (400K vs. 128K) translate to better decision-making, and whether model architecture matters more than context capacity.]

2.5.7 Error Analysis

[TODO: Summary of error patterns—

- Action category confusion matrix (which types are most often confused?)
- Temporal sensitivity (does accuracy degrade deeper into a visit or later in a trajectory?)
- Patient complexity correlation (does performance correlate with visit count or event density?)
- Clinically reasonable alternatives (among incorrect EM predictions, what fraction represent clinically defensible actions?)

]

2.5.8 Summary of Findings

The experimental results converge on a coherent narrative: (1) Clinical decision-making remains challenging (accuracy 0.55–0.62 across all models); (2) Memory granularity matters modestly (Event slightly outperforms Note); (3) Memory augmentation (RAG, Mem0) does not significantly improve decision-making—the central negative result; (4) Immediate context dominates historical context; (5) Chain-of-thought reasoning can degrade performance through increased conservatism. These findings collectively reveal a retrieval-reasoning gap whose implications we explore in Chapter 3.

2.5.9 Qualitative Case Study

To illustrate the decision-making task concretely, consider a representative patient from our cohort: a 68-year-old male with chronic kidney disease, type 2 diabetes, and hypertension, with 21 hospitalizations over five years. At a mid-visit decision point during his 15th admission (presenting with fever and elevated creatinine), the agent receives:

- **Contextual Memory:** Admission event (fever, hypotension), initial lab results (elevated WBC, creatinine 2.8 mg/dL), and one prior imaging order (chest X-ray, pending result).
- **Historical Memory (Event):** Events showing that during prior febrile episodes, the patient responded to vancomycin but developed acute kidney injury requiring dose adjustment; a CT abdomen from visit 12 revealed nephrolithiasis.
- **Decision Query:** "Based on the patient's history and current status, what should be the next clinical action?"

The ground-truth action (from actual records) is ordering blood cultures and starting empiric vancomycin at renally adjusted dosing. An agent relying solely on Contextual Memory might order broad-spectrum antibiotics at standard dosing; an agent effectively integrating history would recognize the need for renal adjustment based on prior vancomycin-associated AKI and the current elevated creatinine. This case illustrates the type of decision where historical integration should change the answer—and where our benchmark can measure whether it does.

2.5.10 Discussion of Negative Results

The finding that memory augmentation does not improve decision-making may appear discouraging, but it carries productive implications. First, it identifies a genuine capability gap that current architectures cannot bridge through scale or retrieval quality alone—the gap is architectural, not parametric. Second, it provides a clear target for future work: architectures that explicitly model the relationship between retrieved history and current decisions, rather than treating retrieval as an independent preprocessing step. Third, it validates LongMedBench's sensitivity: a benchmark that showed all approaches performing equally well would be uninformative; one that reveals systematic failure modes provides actionable diagnostic signal.

The contrast with EHR-Bench [13] is instructive. While EHR-Bench demonstrated that LLMs can predict next clinical events from complete histories with non-trivial accuracy, our agent-based setting reveals that this capability does not translate to interactive decision-making where history must be selectively retrieved and integrated. The two paradigms evaluate complementary but distinct capabilities—static prediction vs. interactive planning—and gap between them highlights the importance of evaluation methodology in assessing clinical AI readiness.

Chapter 3. Discussion and Conclusion

3.1 Summary of Findings

The experimental results presented in Chapter 2 yield five key findings:

1. Decision-making remains challenging. Across GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo, T3-D accuracy ranges from 0.55–0.62 (Exact Match), substantially below clinical deployment requirements. The consistency of this ceiling across architectures suggests a fundamental limitation of current LLM approaches rather than model-specific weakness.

2. Memory granularity matters modestly. Structured Event Memory slightly outperforms Note Memory in most conditions, but the effect is small relative to the overall performance ceiling. For proactive clinical planning, immediate context dominates.

3. Memory augmentation does not improve decision-making. This is the central finding. Neither RAG nor Mem0 provides significant improvement over the naive long-context baseline. In some conditions, augmentation slightly degrades performance.

4. Immediate context dominates historical context. The $n = 0$ baseline (Contextual Memory only, no history) achieves accuracy comparable to full-history conditions. Adding historical visits does not monotonically improve decisions.

5. Chain-of-thought can degrade performance. For DeepSeek-v3.2, reasoning mode reduces accuracy from 0.607 to 0.593, likely because explicit reasoning induces conservatism that penalizes decisive action under exact-match evaluation.

3.2 The Retrieval-Reasoning Gap

These findings collectively reveal a **retrieval-reasoning gap**: a disconnect between LLMs' ability to locate relevant historical information and their ability to integrate that information into proactive clinical planning. This explains why RAG and Mem0 fail: these architectures optimize retrieval quality, but for clinical decision-making, retrieval is not the bottleneck. The model can access relevant history (either in full context or via retrieval), but cannot effectively use it to inform decisions.

We hypothesize three contributing mechanisms. First, clinical reasoning requires causal, not associative, inference—determining whether a prior treatment was effective requires understanding why it was given and how the patient responded, not just retrieving that it occurred. Second, the signal-to-noise ratio in longitudinal EHR data is low: a patient's 900 events contain only a fraction relevant to any single decision, overwhelming attention mechanisms. Third, current LLM training data (web text, code, books) lacks the temporal-causal structure of clinical trajectories, creating a distribution shift.

3.3 Implications

3.3.1 For Memory Architecture Design

Current memory architectures (RAG, Mem0, MemGPT) operate on the assumption that better retrieval yields better performance. Our findings challenge this assumption for clinical decision-making. Future architectures should emphasize <u>reasoning over retrieved information</u>: temporal attention mechanisms that encode causal relevance, graph-based representations that capture clinical relationships, and structured reasoning layers that bridge retrieval and decision.

3.3.2 For Benchmark Design

If decision-making performance is dominated by immediate context, benchmarks that primarily test retrieval may overestimate clinical readiness. We advocate for adversarial task design—constructing decision instances where history <u>changes</u> the correct answer—to prevent models from succeeding through immediate-context heuristics alone. Additionally, distinguishing the subset of decisions where history genuinely matters from those driven by current presentation is critical for targeted evaluation.

3.3.3 For Clinical AI Deployment

The finding that immediate context dominates is not necessarily a weakness—in many clinical scenarios, the current presentation is indeed the most important determinant. The challenge lies in identifying decisions where ignoring history constitutes clinical error (e.g., prescribing a medication that previously caused an adverse reaction) and ensuring the agent has the architecture to recognize and act on such cases.

3.4 Limitations

Data Scope. LongMedBench is constructed from MIMIC-IV, a single-institution database. Clinical practice patterns and documentation conventions vary across institutions, and findings may not generalize without cross-institutional validation.

Cohort Bias. The ≥ 15 hospitalization filter selects for high-acuity, chronic, multimorbid patients—the population for whom longitudinal reasoning matters most, but not representative of the broader patient base.

Task Simplification. T3-D reduces clinical decision-making to action-type prediction within a predefined action space. Real decisions involve drug dosing, contraindication checking, patient preferences, resource availability, and guideline compliance not captured by our formulation.

Evaluation Rigor. Exact Match penalizes clinically reasonable alternatives. While Hit-Any partially addresses this, neither metric replaces clinical expert judgment. We recommend human clinician evaluation as a future validation step.

Model Coverage. Our evaluation covers three general-purpose LLMs. Models specifically trained for clinical tasks (e.g., EHR-R1, Med-PaLM) may exhibit different patterns not captured here.

3.5 Future Work

Task Refinement. Adversarial decision instances where history changes the correct answer would directly test temporal integration. Counterfactual evaluation—altering history and measuring decision sensitivity—would isolate causal effects.

Memory Architecture Development. Temporal graph memory, selective history summarization (compressing irrelevant while preserving salient information), and memory-aware fine-tuning on synthetic longitudinal trajectories are promising directions.

Human Baseline. A clinician evaluation study contextualizing the 0.55–0.62 accuracy range would validate benchmark difficulty and enable calibrated assessment of model progress.

Cross-Institutional Validation. Extending the pipeline to eICU, HiRID [28], or STARR would test generalizability across institutions and patient populations.

Multimodal Extension. Incorporating medical imaging and physiological waveforms would increase clinical fidelity as vision-language and multimodal LLMs mature.

Outcome Correlation. Linking agent decisions to simulated patient outcomes (mortality, readmission, length of stay) would provide the ultimate validation of decision-making quality.

3.6 Conclusion

Clinical practice is inherently longitudinal—patients accumulate histories over years, and clinicians must integrate evidence across this temporal expanse. As LLMs move toward clinical deployment, their capacity for longitudinal reasoning becomes a critical benchmark.

This thesis presented the decision-making component of LongMedBench, investigating how contemporary LLMs use historical clinical information to inform proactive planning through a three-tier memory architecture and systematic ablation experiments. The central finding is that memory augmentation—the dominant paradigm for extending LLM capabilities—does not significantly improve longitudinal clinical decision-making. Decision accuracy depends primarily on immediate context, and historical information, however organized, contributes marginal additional value.

This finding clarifies the nature of the challenge. The bottleneck is not retrieval—finding the right facts—but integration—understanding how those facts collectively inform decisions. Closing this retrieval-reasoning gap requires architectures that go beyond retrieval to support structured reasoning over temporal clinical evidence. The path to clinically capable AI agents is longer than single-encounter benchmarks suggest, but by identifying where current approaches fall short, LongMedBench provides a platform for measuring and ultimately closing this gap.

3.6.1 Relationship to the Broader LongMedBench Project

While this thesis focuses on T3-D (Decision-Making), placing our findings in the context of the broader LongMedBench project reveals complementary insights. The T3-A (Fact-based QA) evaluation found that RAG and agent memory systems significantly improve factual retrieval from long clinical histories [22]. T3-N (Temporal Reasoning) revealed that LLMs struggle with implicit temporal inference (visit ordering without explicit timestamps) despite strong performance when timestamps are provided.

The contrast between these task suites sharpens our central finding: memory augmentation does help for retrieval tasks (T3-A), but does not help for decision-making tasks (T3-D). This dissociation confirms that the retrieval-reasoning gap is task-specific—it manifests when the agent must do more than locate facts, when it must plan proactively. This insight would not be visible from any single task suite; it emerges only from the systematic, multi-dimensional evaluation that LongMedBench provides.

3.6.2 Practical Considerations for Reproducibility

To facilitate reproducibility and future extensions, we release the complete data processing pipeline as open-source software. The pipeline is containerized, with explicit version pins for all dependencies and documented configuration parameters for each stage. The MIMIC-IV data itself requires credentialed access through PhysioNet, but the processing scripts, task generation logic, and evaluation harness are publicly available. We provide a synthetic example dataset with identical schema to enable testing and development without MIMIC-IV access, following the precedent established by EHR-Bench and MedAgentBench.

All experiments were conducted on [TODO: computing platform, GPU specifications]. API-based model evaluation used standardized temperature settings (temperature = 0 for reproducibility) and fixed random seeds where applicable. Full prompt templates, including system prompts, memory formatting conventions, and decision queries, are documented in the released codebase to ensure exact replication.

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Appendix

(if applicable)

You can attach your publication, patent document, etc here.

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(if applicable)

(Main text: Times New Roman, size 12, 1.5 linespace)

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